Virtual Reality-Based Airport Navigation Assistance System Using Signage-Guided Interactive Wayfinding

MTE 4442: Augmented Reality & Virtual Reality

Project Synopsis

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1. Introduction

Airports are large, visually busy environments, and first-time or infrequent travellers often struggle to find their way from the entrance to check-in, through security, to the correct boarding gate, and finally to baggage claim after landing. Confusing signage, long unfamiliar corridors, and the stress of time pressure can lead to missed flights, wrong queues, and general passenger anxiety. Airports cannot easily be used to train or study this kind of wayfinding behaviour in real life, since the terminal is operational, security-sensitive, and not something a passenger can "practice" in ahead of time.

This project proposes a VR Airport Navigation Simulator built using Unity 6.5. The system places a user inside a virtual airport terminal and requires them to move through the standard passenger journey: locating the check-in counter, passing through a security checkpoint, finding the correct boarding gate from a flight information display, and, in a return scenario, retrieving luggage from the correct baggage carousel. Directional signs, gate numbers, and information boards are placed throughout the terminal exactly as they would be in a real airport, and the user must read and interpret them to navigate correctly.

The project will implement the required components of a VR system.

Component	**Implementation**
Virtual World Generator	Terminal layout including check-in hall, security lane, concourse, gate areas, and baggage claim hall
Visual Renderer	Terminal interiors, signage, flight information displays, queues, and lighting
Visual Display	Compatible VR headset
User Interaction	VR movement and interaction with counters, kiosks, scanners, and signs

2. Method

The proposed Simulator will be developed using the XR Interaction ToolKit in Unity 6.5 to make it compatible for Headset Tracking, Controller Input, and Object Interaction. The various stages of the Passenger Journey will be built as four modular zones, each stage being developed and tested before being integrated into the Simulator.

Table 1: Terminal Zones

Zone	**Key Elements**

Check-In Hall	Entrance, Check-In Desks, Signage
Security Checkpoint	Queue Lanes, Screening Point
Departures Corridor	Gate Numbers, Flight Info Displays, Directional Signs
Arrivals/ Baggage Claim	Carousels, Carousel Displays

Development of the Simulator will start from a prototype of the Check-In Hall and then progress through all the stages of the journey, with usability checks at each stage for comfort, scale, and signage clarity.

Upon completion, participants will navigate the full simulation, with their performance being tracked via time required to complete tasks, number of wrong turns, and self-reported anxiety/confidence, alongside qualitative feedback on realism and signage clarity.

3. Experiments

The experimental assessment will gauge the usability of the VR system and examine passenger wayfinding efficiency under different navigation circumstances throughout the virtual terminal. The assessment will include three testing scenarios designed to evaluate how participants read terminal signage and respond to navigation obstacles.

Baseline Wayfinding (Departure Journey)
The participant starts at the entrance of the main terminal and needs to find their specific check-in counter, go through security, and make their way to the designated boarding gate by following the directional signs and the Flight Information Display Systems (FIDS).

Signage Obstruction / Visual Clutter
Signs located above at important decision points (such as concourse junctions) are either partially hidden or visually complicated, which forces users to depend on additional cues or interactive information kiosks.

Arrival & Baggage Retrieval Journey
In order to locate and get to the appropriate baggage claim carousel, the user must follow arrival signage via hallways after spawning at an arrival gate.

Experiment	Condition	Measurements
Standard wayfinding	Full clear signage, standard layout	Total navigation time, path trajectory or distance, number of wrong turns, task success rate
Restricted Signage	Obscured or missing direct signage at key junctions	Time to reorient, decision point hesitation time, kiosk interaction count, error count
Arrival and baggage claim	Multi-corridor route to baggage hall	Route completion time, carousel identification accuracy, self–reported navigation ease

A small group of participants will take part in a pilot test. Data on objective metrics (such as time taken, position logs, and error counts) will be automatically exported to a CSV file. After completing the simulation, participants will fill out a standardized System Usability Scale (SUS) questionnaire and a custom feedback form that assesses the clarity of signage, visual comfort, and their confidence in wayfinding. All final quantitative data will be sourced from testing the finished Unity prototype.

4. Conclusion

The project aims to create a workable VR airport navigation simulator with the help of Unity, incorporating features such as an immersive environment of a virtual terminal, actual navigation signs, virtual movement and interaction features, and the capability of measuring navigation effectiveness through different parts of a terminal experience, including check-in, boarding, security, and baggage claim.

In making the project feasible for the students, it restricts itself to one simple terminal layout and a limited number of gates. Also, it makes the first prototype independent from any VR multi-terminal settings, autonomous agents, multiplayer components, and any connection with real-time airport data.

The final system will demonstrate how VR can be used to study and improve passenger wayfinding in airport terminals, offering a safe, repeatable environment in which signage and layout choices can be tested before being applied in a real airport.

References

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